# Optimal Sizing of a Micro-Reactor for Embedded Grid Systems

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## INTRODUCTION

There has been significant work recently to develop energy system concepts that incorporate a mixture of nuclear energy, variable renewable energy (VRE) and energy storage techniques. These systems are often referred to as nuclear hybrid energy systems (NHES) and are generally robust, reliable, economically appealing, and have low to zero greenhouse gas (GHG) emissions [1, 2, 3, 4]. In 2015, the University of Illinois at Urbana-Champaign (UIUC) made a commitment to reach carbon neutrality by 2050 in the Illinois Climate Action Plan (iCAP) [5]. This plan identified nuclear energy as a potential contributor to this goal. In this work we use the `RAVEN` framework to find the optimal size for a micro-reactor that would minimize the levelized cost of electricity (LCOE) for the UIUC embedded grid. The greatest source economic variability is contained in the capital costs of a micro-reactor. Thus we examine both scenarios where the reactor is provided at no cost to the university and purchased at full price.

## BACKGROUND

UIUC uses steam to heat a majority of it's buildings and chilled water to cool them. The campus power plant, Abbott Power Plant (APP), produces 100% of the campus' steam needs and typically all of the electricity needs. Notable exceptions are peak demand days where the power output of APP, the solar farm, and the wind power purchase agreement (PPA), must be supplemented with electricity purchased from the Mid-continent Independent System Operator (MISO). APP is a co-generation power plant, which means it produces steam and electricity concurrently. In this work we consider only the electricity production. The university has a 10-year PPA with Rail Splitter Wind Farm located in Lincoln, Illinois and a 10-year PPA with Phoenix Solar South Farms, LLC that operates a solar farm on campus [6, 7].

## METHODOLOGY

In this work we use and extend the methodology described in Baker et. al using the `RAVEN` framework [1, 8]. This methodology allows for empirical quantification of energy system flexibility. Baker et. al used a fixed amount of nuclear power production from a small modular reactor (300 MWe) and varied the penetration of VRE in the form of wind power. They also included grid flexibility in the form of a fixed amount of battery storage. UIUC has wind and solar power purchase agreements that fix the amount of VRE the campus receives. Rather than fixing the nuclear power production, this methodology will be extended to search for the optimal size of a micro-reactor for the campus.
There are three main steps to this methodology [1]:

1. Generate synthetic data by training a reduced order model (ROM) with historical data using `RAVEN`'s `TrainARMA` functionality. `RAVEN` condenses historical data into a typical time series and attempts to remove anomalous data.
2. Calculate the net demand for each history set.
3. Pass the net demand to the nuclear hybrid energy system (NHES) dispatch model defined for the UIUC embedded grid.

Net Demand

Similar to Baker et. al, we define the net demand to be

$$D_{net} = (D_{total} - P_{wind} - P_{pv})_h \ \forall \ h \text{ in } [0, 8759] \tag{1}$$

Where $D_{total}$ is the total demand at a given hour, $h$, of a synthetic demand profile from the ARMA model. We adjust the net demand, $D_{net}$, because VRE is used regardless of demand (thus it could exceed demand). $P_{wind}$ is computed by [9]:

$$P_{wind} = \begin{cases} 0 & \text{if } U_h \geq 25 \\ 0.5\eta\rho U_h^3 \frac{\pi d^2}{4} & 3 < U_h \leq 12 \\ 8.6 & 12 < U_h < 25 \end{cases} \tag{2}$$

where

$\eta$ is the conversion efficiency (0.31)
$\rho$ is the density of air at the site
$U$ is the wind speed in [m/s]
$d$ is the diameter of the turbine blades (77 m)

The value for peak output is based on the UIUC wind power purchase agreement [6]. $P_{pv}$ is available as historical data from AlsoEnergy [10]. There are times when the meters failed to record data so power output for those times was calculated by [9]:

$$P_{pv} = G_T \tau_{pv} \eta_{ref} A[1 - \gamma(T - 25)] \tag{3}$$

where

$$G_T = DNI * \cos(\beta + \delta - lat) + DHI * \frac{180 - \beta}{180} \tag{4}$$

$$\delta = 23.44 * \sin\left(\frac{\pi}{180}\frac{360}{365}(N + 284)\right) \tag{5}$$

where

$N =$ day of the year

$DNI =$ Direct Normal Irradiance [kW]

$DHI =$ Diffuse Horizontal Irradiance [kW]

$G_T =$ total irradiance [kW]

$\eta =$ conversion efficiency (0.15)

$\beta =$ tilt of the solar panels [$degrees$]

$\delta =$ declination of the Earth [$degrees$]

$T =$ temperature [°C]

$A =$ area covered by solar farm [$m^2$]

$\gamma =$ temperature coefficient (0.0045)

$\tau =$ transmittance of the PV module

These values were obtained from the iSEE facts sheet about the UIUC solar farm [7].

Dispatch Model

The dispatch model is similar to that used by Baker et. al [1]. The net demand is defined by Eq. 1. The dispatch model first checks if renewables satisfy the campus grid demand. This can never happen with the current installed renewable capacity at UIUC. Then a hypothetical micro-reactor will produce electricity to the grid. Two assumptions will be made: First, the micro-reactor will only produce electricity and not co-generate like APP; second, the micro-reactor will operate at 100% capacity all of the time. This second assumption is a real possibility given the range of sizes we are investigating. If the reactor can meet and exceed the net demand then the dispatch model will attempt to store the surplus in the campus chilled water system. If the chilled water tank is full, then the surplus must be sold back to the grid operator (MISO) and a penalty will be incurred. When the reactor cannot meet the electric demand, APP will produce electricity to fill the difference. UIUC will rely on APP for load following because natural gas boilers can, currently, ramp more quickly than nuclear reactors. If either the demand is not covered or all surplus power is not used then a penalty is incurred.

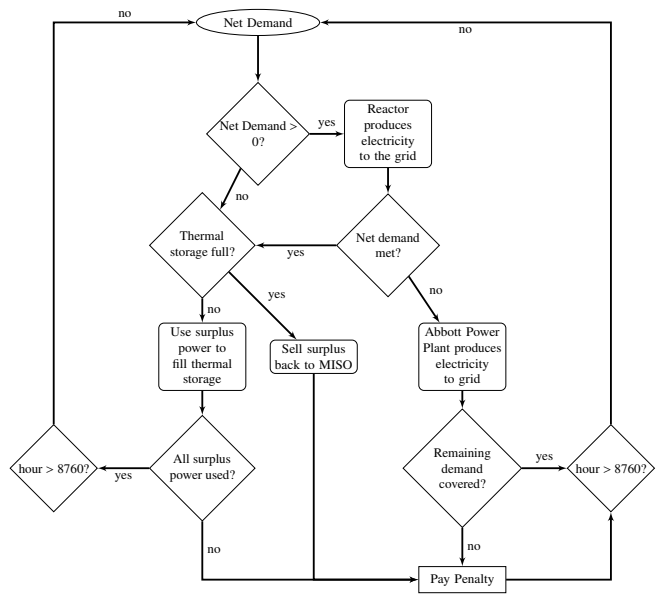


Fig. 1: Dispatch flow diagram for the UIUC NHES

Capital Cost

As in Baker et. al, capital costs are taken in full at year zero. In the first scenario we examine the simplifying assumption is that there are no capital costs because the reactor would be provided at no cost to the university. For the second scenario we will apply the model from Baker et. al. The reference plant used in that work is an 1100 MWe pressurized water reactor (PWR) which costs \$4100/kWh for an ideal case. $x = 0.64$ to model the economies of scale [1].

$$CF^{capital} = \alpha^{year} \left( \frac{P_\mu}{P_{ref}} \right)^x \tag{6}$$

where

$\alpha^{year} =$ construction cost at year 0

$P_\mu =$ size of the micro-reactor

$P_{ref} =$ size of the reference plant

## RESULTS

Current results include typical time series for quantities of interest. Namely steam demand, electric demand, and power produced by VRE. We have also generated synthetic data for grid demand and solar power using the `trainARMA` function in `RAVEN`. Thus the first step of the methodology from Baker et. al has been completed.

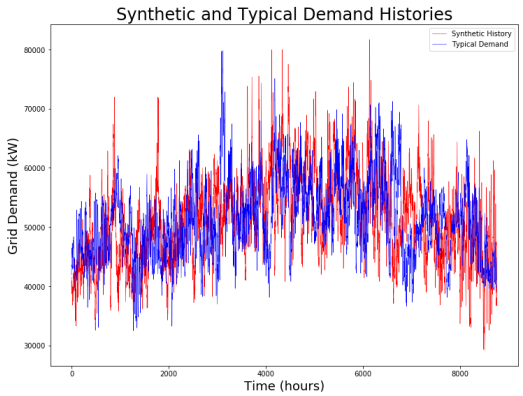


Fig. 2: The typical year of hourly grid demand in kW at UIUC.

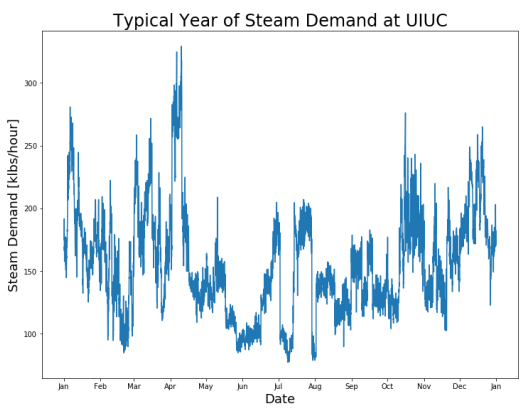


Fig. 3: The typical year of hourly steam demand in kW at UIUC.

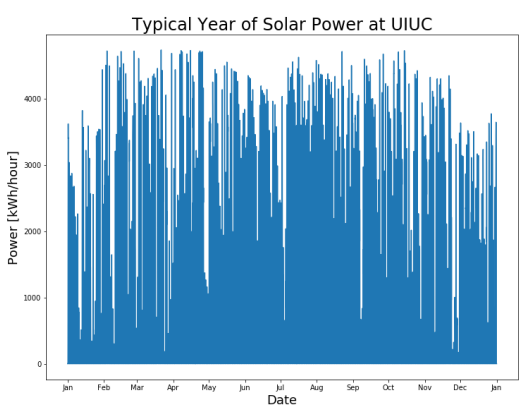


Fig. 4: The typical year of hourly power produced by the UIUC solar farm.

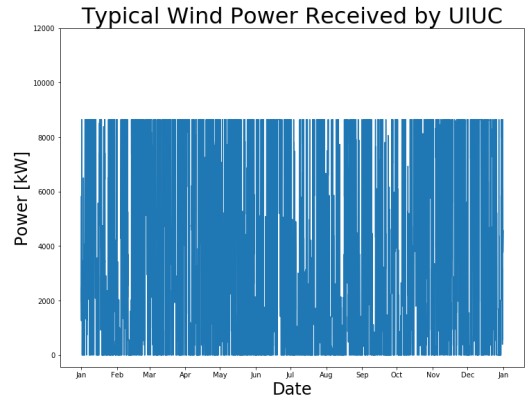


Fig. 5: The typical year of hourly wind power received by UIUC.

## CONCLUSIONS

Baker et. al found that increasing penetration of renewables increased grid volatility which exponentially increased the LCOE of the energy system when subject to a mismatch penalty. They also found that energy storage becomes increasingly valuable for grid stability at or above 20% renewable penetration. The UIUC campus does not yet have enough renewable capacity to meet this criterion. It is also critical to note that in this work we largely ignored the steam demand for the UIUC campus despite its significance to the overall energy system. Future work will consider the role of a micro-reactor on campus where it is enabled for co-generation and can replace natural gas capacity at APP. Here we have shown that the methodology outlined in [1] can be extended to another energy system.

## ACKNOWLEDGMENTS

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